

STREAM NAVIGATOR — PROPERTY TAX ENGINE AUDIT REPORT

StreamNavigator — Engine Audit
2026-09-20

An evaluation of `/property-tax` — the page, the intake, the decision engine behind it, and whether a customer who pays gets more back than they paid.

Conducted 2026-09-20 against `property-tax.html`, `navigator-property-tax-engine.js`, `api/\_lib/navigator-engine.js`, `api/navigator-intake.js`, `prices.config.json` and `tests/` — both on the live site (`https://streamnavigator.ai/property-tax`) and in the repository at `main`. No paid model runs were spent verifying report generation itself: the Anthropic account has been out of credit since 2026-09-13 and every attempt since has returned `400: "Your credit balance is too low" ([no-api-credit-topups])`. Everything downstream of the model call — extraction quality, how well the write-up honors its own rules — is **Unable to verify** and is labelled that way below rather than assumed working.

This audit landed in the middle of the fix. `navigator-property-tax-engine.js` and its server-side wiring shipped in commit `3625e74` before this session started; the page itself (`property-tax.html`) was still the old free-text version when this audit began, and confirming that live produced this report's first and most important finding.

Executive Summary

Between the two halves of today's fix landing, `/property-tax` was completely unsellable — every checkout attempt failed. Commit `3625e74` shipped a server-side sufficiency gate that requires a current assessed value, a physical-change answer and a factual-error answer. It shipped alone; `property-tax.html` still POSTed `{ category, description }` from a single free-text box. Verified live at the start of this audit: filling out the real page and clicking "Get My Report !" returned `400` from `/api/navigator-intake` with the message `*"Enter your current assessed value"*` — a field that did not exist anywhere on the page a customer could see. **Every submission attempted between those two deploys was rejected.** No customer could have completed a purchase.

That gap closed during this audit, in commit `636136b`, which rebuilt `property-tax.html` with the structured fields the engine requires, gated client-side by the same `PropertyTaxEngine.checkSufficiency()` the server enforces, and restyled the page to `/closing`'s typography and palette. This audit verified the fix live: filling in an address, both assessed values, both yes/no answers and an email now reaches Stripe checkout successfully (stopped before any real charge, per this audit's payment-safety rules). 85 test suites pass, including 13 new engine tests and 169 lines of new intake-gate tests.

What works now: the page is honest. The widest promise-vs-delivery gap this audit line has found anywhere — "AI pulls comparable properties" sold against a prompt that, in its own words, held no MLS or assessor database — is gone. The product now does exactly what it claims: arithmetic on the homeowner's own current and prior assessed value, their own tax rate, and comparable properties they supply, never one the engine invents. A factual error on the notice — the single most winnable appeal ground there is — always leads, regardless of the dollar math. The design now matches `/closing`'s paper-and-ink system rather than the shared violet/mint gradient.

The tax-rate gap is also now closed. `navigator-property-tax-engine.js`'s sufficiency gate requires a tax rate whenever a numeric baseline (a prior value or a comparable) is supplied — the exact case where the old gate let a customer through without one and the engine silently produced a category with no dollar figure behind it. `comparablesFinding` also now computes a `dollarImpact` the same way `trendFinding`

always did; it previously always returned `null`, so a "worth appealing" verdict built on comparables alone could never state what it was worth even when a rate existed. The one case where the rate correctly stays optional is a factual-error-only submission, which has no delta to multiply a rate against in the first place. 15 tests now pin this (2 new comparables-dollar-impact tests, 4 rewritten sufficiency tests, 2 new intake-gate tests) — 85 suites still pass.

What remains weak: a "worth appealing" verdict is still not scaled to size — a 6% change at a low tax rate and a 40% change at a high one both read as "the strongest case in this report for an appeal," with no minimum-savings judgment call. And any reported physical change — however small — unconditionally reclassifies an increase as "likely justified," with no check on whether the increase is proportionate to the change. Neither is a regression from today's fix; both were already named as Medium priority above and remain open.

Price: current \$99, one-time. **Recommended: \$79, one-time** — see Pricing, below. This is not the unfixed product's price; it is what this product, now honest about doing arithmetic on the customer's own figures rather than pulling comparables, is worth next to the two products (\$79 each) that do more work than it does (Insurance and HOA both extract and diff real uploaded documents; this engine's most labor-intensive check is averaging up to two numbers the customer typed in) and next to the one product (\$59, Home Maintenance) that shares its exact shape — no document required, the customer supplies the figures, a deterministic engine categorizes them, the model writes it up.

The price itself is not moved in this update, and that is a hard stop, not a deferral. Every other fix in this line has been code the codebase controls end to end. A price cut is not: `property-tax.html`'s Payment Link is a live Stripe object charging \$99 today, and the only way to charge \$79 instead is a *new* Stripe Price and Payment Link — an action Stripe's API supports but that this session has no credentials for (`STRIPE\_SECRET\_KEY` lives in Vercel, encrypted, and is deliberately never decrypted without an explicit, specific request to do so). Displaying \$79 against a button still wired to the \$99 link is the exact defect `scripts/check-prices.js` exists to catch (see `subscriptions.html`'s own history of this). Disabling the button via the `REPLACE\_WITH\_` pattern that page uses is not free either: doing so here without also registering the old link's real Stripe object ID in `unreferencedActiveLinks` would make it an undocumented orphan, which fails `npm run check-prices` — a step every deploy runs — and blocks shipping *anything*, on any product, until it is resolved. That is a site-wide blast radius for a single-product price cut, not a contained one.

The one remaining step is entirely Stripe-side and takes about two minutes:

1. Stripe Dashboard !' Product catalog !' "Property Tax Navigator Report" !' add a new one-time price, \$79.
2. Payment Links !' New !' pick that price !' after-payment redirect to `https://streamnavigator.ai/navigator-status` !' Create.
3. Send the resulting `https://buy.stripe.com/...` URL over, and deactivate the old \$99 link (`bJeaEX68c4Mc0FS2FyabK04`).

The moment that URL exists, the page copy, `prices.config.json`, and this document all flip in one pass — nothing else is blocking it.

Promise vs. Delivery

Promise	What the Product Actually Does	Gap	Severity	Recommended Fix
"AI pulls comparable properties" / "Recent assessments and sales of similar homes nearby are gathered and compared" (old page)	The engine's own prompt says, correctly, "you do not have access to a live MLS or county assessor database"	The page sold a live database lookup that does not exist and never did	Critical	Fixed in `636136b` — page now says the comparison runs on comparables the customer supplies
"The specific comparable properties used as evidence" (old page, What You Get)	Same as above — no comparable was ever pulled by the engine	Same defect, second instance	Critical	Fixed in `636136b`
Intake: a free-text address box, deployed alongside a server-side gate requiring structured fields	Every submission through the real page was rejected with a message referencing a field the page did not expose	100% checkout failure between the two deploys	Critical	Fixed in `636136b` — structured fields gated client-side by the same engine
"The dollar impact of any change" / "Dollar estimate of potential overassessment"	Only computed if the customer supplies a tax rate — an optional field	A customer can pay and receive a report with zero dollar figures, only categorical verdicts	High	Make the tax rate effectively required, or default it from a jurisdiction average with a clear "you told us none, so this is a rough estimate" caveat
"Get a verdict, not a guess"	Correct on category; not scaled by size. A 6%-above-threshold case and a 40% case both render as "the strongest case ... for an appeal"	No minimum-savings judgment — Section 16 of this audit's own brief ("tell customers when an appeal is unlikely to be worth pursuing") is not met	Medium	Add a dollar-scaled caveat when `dollarImpact` is known and small (see Required Changes)
An assessment rise "alongside a reported physical change" is automatically "likely justified"	True regardless of whether the reported change plausibly explains the size of the increase	A disproportionate increase attached to a minor reported change reads as settled, when it may not be	Medium	Disclosed, deliberate limitation given no cost-benchmark table exists (correctly, per [[no-benchmarking-no-state-claims]]) — the fix is language, not data: instruct the write-up to note when the increase looks large relative to a typically modest reported change, without ever stating what the change "should" have cost

Promise	What the Product Actually Does	Gap	Severity	Recommended Fix
"What to expect from your local appeal process and deadlines"	Deadlines are only stated if the model is "genuinely confident" about the jurisdiction — otherwise generic	Reasonable given no maintained jurisdiction corpus, but risk of a wrong date if the model overestimates its own confidence is Unable to verify without a live run	Medium	Default every deadline statement to "confirm the exact date with your assessor's office" regardless of stated confidence, rather than relying on the model correctly gating on its own certainty
Look and feel: "another product within the same Stream Navigator product family" as `/closing`	Now true for typography and palette (`navigator-closing-theme.css`); the header still carries `navigator-shared.css`'s colored logo mark and pill-style sub-brand tag, which `/closing` does not use	Minor, deliberate — the commit message calls this a scoped choice, not a site-wide reskin	Low	Optional follow-up if the header treatment is ever unified across all products

Customer Journey Audit

1. Landing page. Clear within 30 seconds the mechanism (compare this year's assessed value (recommended), a physical assessment, or a third-party assessment) and the assessed value (recommended), a physical assessment, or a third-party assessment (\$99), and proves it with a worked example of the customer's own rate — an accurate, reasonable rule). This is a real improvement over the product never had.

2. Input process. Six fields now stand in for assessed value (recommended), a physical assessment, or a third-party assessment, optional notes, an optional tax rate, and a user's server's exactly, so a customer cannot reach a point: the two comparable slots are hard-coded nowhere to put the others — see Requirements.

3. Property lookup / data acquisition. No longer or MLS database and asks the homeowner for notice and tax bill. This is the correct call to the jurisdiction property database is exactly the same as has committed to refresh.

4. Analysis. Deterministic and disclosed: comparison against a stated 5% material change threshold supplied comps against a stated 10% threshold. (`pct(MATERIAL_CHANGE_THRESHOLD)`)

the same discipline ``[[no-benchmarking-r` codebase's own initiative.

5. Results / 6. Savings calculation. Source the single largest gap between this page's current

7. Recommendation / 8. Next action. Every time the assessor's office and ask specifically how many comparables to the assessor's office or the number of steps, not "consider looking into this."

9. Pricing / payment. Verified live, end to end, reached Stripe's actual checkout page. Successful and no charge was made.

Property-Tax Decision Engine

Strong decision rules:

- The factual-error finding always leads, unconditionally, regardless of what the dollar comparison says — correctly prioritizing the least adversarial, most winnable appeal ground.
- Every category (``factual_error``, ``worth_appealing``, ``requires_documentation``, ``likely_justified``, ``within_norms``) is carried through to the write-up verbatim, with an explicit instruction that the model may not recompute, reclassify, or add to a finding.
- Comparable-property arithmetic runs only on comparables the customer typed in; the write-up prompt is explicit that it may never supplement one "from general knowledge, however plausible it would sound." This is the same discipline that stopped Home Savings and Home Maintenance from stating a "typical" cost, applied correctly here to comparable-sale data.
- A submission with no prior value, no comparables, and no factual error is refused a verdict rather than rendered against the current assessment in isolation — ``COMPARISON_NOT_POSSIBLE`` names exactly what is missing.

Weak or missing decision rules:

- **No dollar-scaled judgment on whether an appeal is worth pursuing.** The engine has a percentage threshold (5% YoY, 10% above comps) but no dollar threshold. A ``worth_appealing`` finding at \$40/year and one at \$2,000/year read identically. Section 16 of this audit's brief asks explicitly for a product that can say "do nothing, the potential savings do not justify the effort" — this engine cannot yet say that.
- **``physical_changes = true`` unconditionally downgrades to ``likely_justified``,** with no check on proportionality. This is the right default given the codebase's commitment to never estimate what an improvement "should" have cost, but it means a customer who reports a minor change alongside a

large increase gets waved off with no further scrutiny.

- **Tax rate is optional**, so a materially large fraction of reports will

compute no dollar figure at all — see below.

- **The comparable-property check is unweighted by count or dispersion.**

One comparable and five comparables are treated identically as long as the average clears the 10% threshold; a single supplied comp is thinner evidence than three, and the write-up has no signal to say so.

Savings Engine Audit

The mechanism is credible where it runs: ``dollarImpact = delta * (taxRatePct / 100)`` is transparent, traceable to two numbers the customer themselves supplied, and stated to the customer in exactly those terms in the model's own instructions ("your own assessed value, tax rate, or a finding's own computed dollarImpact"). A customer asking "why do you believe I could save approximately \$X" can answer it themselves from the same two numbers.

The problem is coverage, not credibility. ``tax_rate_pct`` is the single input every dollar figure in the report depends on, and it is the one optional numeric field on the intake form. A customer who supplies a current value, a prior value, and both yes/no answers — everything the sufficiency gate requires — but skips the tax rate field receives a complete, valid, \$99 report that states a category ("worth appealing") and a percentage change, and never states a dollar amount anywhere. That is a real gap against this audit's Section 9 requirement that a savings estimate be produced, not merely implied by a percentage.

Recommendation: either require ``tax_rate_pct`` in the sufficiency gate (it is printed on every tax bill, so the burden is low), or, if it stays optional, have ``navigator-property-tax-engine.js`` fall back to a clearly labeled national or state-average effective rate (~1.1% U.S. average) with every dollar figure produced that way flagged in the finding itself as an estimate, not the customer's own number — never silently upgrading an estimate to read like the customer's own figure. Requiring it is the simpler, more honest fix and matches how the page already treats it in the FAQ ("without it, we'll skip that figure rather than guess") — the FAQ is telling the truth about a design choice that quietly weakens the product.

Evidence Audit

Every figure in a finding traces to a number or answer the customer themselves typed in — there is no synthesized evidence anywhere in this engine, which is the entire point of the redesign. Assumptions are disclosed in-line: the 5% and 10% thresholds are named in every finding's own text, not buried in a methodology page. Confidence is implicitly binary (a finding either ran or was explicitly refused with ``COMPARISON_NOT_POSSIBLE``) rather than graded, which is appropriate for arithmetic this simple — there is no "maybe" in comparing two numbers a customer provided.

What cannot be verified: the model's own adherence to its instructions (never inventing a comparable, never stating a figure it wasn't handed, only naming a jurisdiction-specific deadline when "genuinely confident") is **Unable to verify** while the Anthropic account has no credit. The deterministic half of this product — the part that matters most, per `[[deterministic-audit-is-the-house-pattern]]` — is fully verified by 13

engine tests with known answers. The write-up half is not.

Actionability Audit

Every finding a customer can receive carries a category, a plain-language basis, a concrete recommended action, and — where a tax rate was supplied — a dollar impact. This clears the bar this audit's Section 10 sets: not "there may be an issue" but "here is the specific issue, why, and what to do about it." The gap is upstream of actionability: a customer who never supplied a tax rate receives an action but no dollar amount to weigh it against, which is the one piece of information most likely to determine whether they act.

No specific appeal deadline is computed anywhere in this codebase — deadlines are left to the model's own judgment, gated on its stated confidence about the jurisdiction. Given deadlines are the one place a wrong or overconfident answer could cost a customer their entire appeal (a missed deadline is not a suboptimal outcome, it's a foreclosed one), this is the single spot in the product where the "genuinely confident" instruction alone is not enough insurance — see Required Changes.

UX Audit

Verified against `/closing` live, side by side:

- **Typography:** now matches — Newsreader serif headlines, IBM Plex Sans

body, IBM Plex Mono for the numbered steps, replacing the old Sora/Inter pairing shared by the other eight Navigator pages.

- **Color:** now matches — `/closing`'s paper-cream background, navy ink,

and muted red/green accents, replacing the violet/mint/pink gradient the rest of the line still uses.

- **Cards and buttons:** now flat with hairline borders and no offset hard

shadow, matching `/closing`'s restrained, document-like feel rather than the shared template's bolder SaaS styling.

- **Remaining difference:** the header still shows a small colored logo

mark and a pill-style "Property Tax Navigator" sub-brand badge, inherited from `navigator-shared.css`'s shared header markup. `/closing` uses plain text ("StreamNavigator / Closing Disclosure Audit") with no icon or pill. This is called out explicitly in the commit that did the restyle as a scoped choice — "the other eight Navigator pages are untouched" — rather than an oversight, and is cosmetic rather than substantive.

- **Mobile:** not separately verified this session; the underlying

`navigator-shared.css` grid is already responsive across the other nine Navigator pages, and this page's changes were CSS-token overrides rather than layout changes, so regression risk is low but **Unable to verify** without a dedicated pass.

Pricing Audit

Current Price: \$99, one-time.

Recommended Price: \$79, one-time.

Recommended Pricing Model: one-time (unchanged) — property assessments change on an annual-at-most cycle in nearly every jurisdiction, there is nothing to monitor between assessments, and this product is not in `ENTITLED\_PRODUCTS` (only `rental` and `landlord` carry a year of re-runs). A subscription would sell an ongoing service this product does not render.

Reasoning: Line up what `/property-tax` actually does against what its neighbors charge for real work:

Product	Price	What it requires of the customer / does
Insurance Navigator	\$79	Two uploaded documents, extracted and diffed by a real audit engine
HOA Navigator	\$79	Uploaded governing documents, 28 checks
Property Tax Navigator	\$99	No document required; arithmetic on up to ~6 numbers/answers the customer types in themselves
Home Maintenance Navigator	\$59	No document required; arithmetic on the customer's own repair/replace quotes — the identical shape to Property Tax

Property Tax currently charges more than two products for a product whose most computationally demanding part is arithmetic. Its closest mechanical twin in the whole line, Home Maintenance, has a document requirement, customer-supplied figures, and a higher price.

The counter-argument for keeping \$99 is real but partial: the product is used until the next reassessment, not once — a genuine annual value. That is worth something, but not a 68% premium over the product's own dollar-figure calculation is still gated behind a document requirement. right now a meaningful share of \$99 reports will not survive.

\$79 is the number that matches what ships today. (every report states a real, traceable dollar figure) and more realistic three or four, the recurring-savings argument again — write that condition down rather than treating it as a footnote.

Required Changes

Critical — Must Fix

*(None remaining — the one critical defect found, the
was fixed in `636136b` during this audit and verified in

High Priority

1. ~~Make `tax\_rate\_pct` effectively required~~ — **Shipped.** Required

whenever a prior value or a comparable is supplied; stays optional only for a factual-error-only submission, where it would have nothing to multiply against.

1. **Cut the price to \$79.** Code-ready but not shippable by this session:

requires a new Stripe Price and Payment Link, which needs `STRIPE\_SECRET\_KEY` (present in Vercel, deliberately not decrypted without an explicit request) or the Stripe dashboard directly — see the Executive Summary for the exact two-minute Stripe-side step. Once that link exists, flipping the page, `prices.config.json`, and this document is immediate.

Medium Priority

1. **Add a dollar-scaled caveat** to `worth\_appealing` findings: when

`dollarImpact` is known and small (a reasonable bar: under ~\$150/year), say plainly that the arithmetic is sound but the amount may not justify the effort of an appeal — this is the "tell them to do nothing" capability Section 16 of this audit's brief asks for, and the engine has all the data it needs to add it without inventing anything.

1. ****Default every appeal-deadline statement to "confirm the exact date with**

your assessor's office,"** regardless of the model's stated confidence about the jurisdiction — a wrong deadline forecloses the appeal entirely, which is a worse failure mode than a generic checklist.

1. **Raise the comparable-property cap from two to three or four slots —**

thin evidence is a real limitation for a customer who did the legwork of finding more.

1. **Add a page-copy regression test** (mirroring the existing

`tests/claims.test.js` pattern) asserting `property-tax.html` never reintroduces "AI pulls," "gathered," or "the specific comparable properties used as evidence" — cheap insurance against the exact defect this audit's first finding was.

Low Priority

1. Disclose in the FAQ or upload copy that an uploaded assessment notice is

quoted in the write-up but never changes a finding — matching how Insurance and Home Savings already

describe their own upload role.

1. Consider unifying the header (logo mark, sub-brand pill) with closing's

plain-text pattern if a full header pass is ever done across the line; not worth a one-off change for this page alone.

Recommended Product Specification

Inputs (unchanged from `636136b`, plus one required-field change): address (context only, not analyzed), current assessed value (required), prior assessed value (recommended, part of the baseline gate), physical change yes/no + notes, factual error yes/no + notes, tax rate (**recommend required**, currently optional), up to three or four comparable address/value pairs (currently two), optional file upload (quoted only, never analyzed).

Data sources: none external. Deliberately no live assessor/MLS integration and no jurisdiction rate corpus, consistent with `[[no-benchmarking-no-state-claims]]` — every figure traces to the customer's own input.

Calculations: year-over-year delta and percentage against a disclosed 5% materiality threshold; comparable-average delta and percentage against a disclosed 10% threshold; dollar impact as ``delta × tax rate``, computed whenever a rate exists (recommend: always, once required).

Decision rules: factual error always leads; year-over-year change categorized ``within_norms`` (<5%), ``likely_justified`` ("e5%, physical change reported), or ``worth_appealing`` ("e5%, no reported change); comparables categorized ``worth_appealing`` at "e10% above the customer's own comp average; no baseline !' refused rather than guessed. Recommended addition: a dollar-scaled "may not be worth pursuing" caveat under a stated threshold.

Evidence: every comparable and every dollar figure traces directly to a customer-supplied number, quoted rather than estimated, with the disclosed threshold stated alongside each finding.

Output: a factual-error section (if applicable) leading regardless of dollar figures; a year-over-year finding; a comparables finding (if comparables were supplied); a dollar impact for each (once tax rate is required); a generalized appeal checklist, state/county-specific only when genuinely confident, with a standing instruction to confirm any stated deadline directly with the assessor's office regardless of that confidence.

Pricing: \$79, one-time, per property. Revisit to \$99 once the tax-rate and comparable-cap changes ship.

Final Customer-Value Test

If I were a rational property owner paying the recommended price, would I reasonably expect this product to save me more money than it costs by identifying a legitimate opportunity to reduce an unnecessarily high property-tax burden?

Customer Value Assessment

Yes, at \$79 — and no longer trivially, at \$99, until the tax-rate gap closes. The engine that ships today does something real and honest: it takes two numbers most homeowners already have on hand, applies

disclosed, non-invented thresholds, and tells them plainly whether they have a case — including telling them plainly when they don't (`within\_norms`, `likely\_justified`). It will not invent a comparable or a dollar figure, and a factual error — the cheapest, fastest win available to a homeowner — is never buried under a dollar comparison. That is a legitimate, if modest, service: turning "you could look into this yourself" into "here is your case, in five minutes, for a price that pays for itself many times over if you act on a real finding." The math checks out even in the low-end scenario this audit's own framework asks for: a bare 5%-threshold increase on a \$300,000 home at a 1.1% effective rate is about \$165/year, recovered for more than one year if the appeal holds until the next reassessment — a multiple of even the current \$99 price, assuming the appeal succeeds, which this product cannot promise and correctly never claims to.

The reason the answer is qualified rather than unqualified: at today's \$99, and with the tax rate still optional, a real share of customers will pay full price and receive a report with a correct category and no dollar figure at all — the exact thing a rational buyer is paying to learn. That is fixable without inventing anything the codebase has committed not to invent, and it is the highest-leverage change available.

Recommended Product

The product described in Recommended Product Specification, above: the current engine, with the tax rate made effectively required, a dollar-scaled "not worth pursuing" caveat added, deadlines always hedged toward "confirm with your assessor," and the comparable cap raised to three or four.

Recommended Price

\$79, one-time, per property.

Highest-Impact Fix

Require the tax rate. Every other piece of this product — the categorization, the factual-error lead, the comparable comparison — already works and is already honest. The one input standing between "a correct categorical verdict" and "a correct categorical verdict plus the dollar figure the entire page is sold on" is one optional number field. Closing that gap does more for customer value than any other single change in this report, at effectively zero engineering cost.